# Akoizo Original Practice - Form 4 - Answer key

### Division B | Boomilever | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

### ANSWER KEY - Review only after submitting.

#### 1. (2 points) Which material is NOT allowed in a Boomilever?

Unit: Wood selection, density, and buckling.
Worked-case data: A strut buckles at 120 N with an unbraced length of 30 cm. A brace is added at its midpoint. Estimate the new buckling load.
Separate investigation observations:
Stick A: 0.11 g/cm^3 - failed as a strut at 6 kg.
Stick B: 0.18 g/cm^3, same size - held 11 kg as a strut.
Stick C: 0.11 g/cm^3 with a midpoint brace - held 15 kg.

A. Basswood

B. Hard balsa

C. Commercial plywood

D. Team-laminated balsa

**Correct: C**

#### 2. (2 points) How do compression members usually fail in a Boomilever?

Unit: Statics of a cantilever: moments, tension, and compression.
Worked-case data: A 15 kg load hangs 42 cm from the wall. The strut meets the wall 20 cm below the bolt. Estimate the horizontal chord force.
Separate investigation observations:
Test A: Strut 15 cm deep, 3 mm square: bowed sideways at 9 kg.
Test B: Same strut with one mid-length brace: held 15 kg.
Test C: Chord 1.5 x 3 mm: never failed in any test.
Test D: Chord-to-bolt joint glued with a single drop: pulled off at 12 kg.

A. By crushing the bolt

B. By buckling sideways

C. By stretching apart

D. By melting the glue

**Correct: B**

**3. (2 points) A joint failed and glue remained on only one surface. What does that suggest?**

Unit: Testing, failure analysis, and iteration.
Worked-case data: Your log shows: 9.2 g held 15 kg; 8.4 g held 15 kg; 7.9 g failed at 13.1 kg (strut buckled). What should the competition build be?
Separate investigation observations:
Builds 1-3: 8.1 g, 8.3 g, 8.2 g: all held 15 kg.
Builds 4-5: 7.4 g, 7.5 g: failed at 12 kg and 14 kg, strut buckled.
Build 6: 7.6 g with a stiffer, denser strut: held 15 kg.

A. Poor bond area or fit at that joint

B. The wood was too dense

C. The load was too far out

D. The bolt was loose

**Correct: A**

**4. (2 points) Why do soft-wood bolt hooks often fail by crushing?**

Unit: Joints, adhesives, and the bolt hook.
Worked-case data: A chord joint must carry 300 N. Your glue holds about 2 N per mm^2 of long-grain overlap. How much bonded area do you need, with a safety factor of 1.5?
Separate investigation observations:
Failure 1: Hook crushed into an oval around the bolt at 10 kg.
Failure 2: Strut-to-chord butt joint split at 13 kg.
Fix trial: Basswood bearing pad (+0.15 g) and two gussets (+0.10 g): held 15 kg.

A. The round bolt concentrates force on a small area of soft wood

B. The bolt heats up under load

C. Glue cannot touch metal

D. The bolt is too far from the wall

**Correct: A**

**5. (4 points) Explain why the bonus contact line usually makes a Boomilever heavier.**

answer: A shallower contact shortens the lever arm of the couple, so chord and strut forces rise, and a flatter strut buckles more easily, requiring more wood or bracing. (4 points).

**6. (2 points) Which prototype should the team bring, based on expected score?**

Supplied synthetic observations:
Prototype 1 (base): 9.6 g, held 15,000 g, contact at 14 cm below the bolt.
Prototype 2 (bonus): 12.1 g, held 15,000 g, contact above the 15 cm line.
Prototype 3 (bonus, lightened): 10.4 g, failed at 12,300 g when a joint cracked.
Supervisor note: Prototype 1's contact point is above the 20 cm line.

A. Prototype 3

B. Prototype 2

C. Prototype 1

**Correct: B**

**7. (4 points) Why is pouring sand steadily better than dumping it quickly?**

Unit: The test apparatus, setup, and loading procedure.
Worked-case data: Plan the six-minute window for a team that has practiced setup.
Separate investigation observations:
Run 1: Setup took 5 min; fast pour; failed at 9 kg with violent swing.
Run 2: Setup took 2 min; slow pour with stick tips; held 15 kg.
Inspection: Both structures were identical builds from the same wood batch.

answer: Fast pouring creates sudden dynamic loads and bucket swing that can exceed the static load and break the structure early. (4 points).

**8. (4 points) Give one reason the tension chord often fails at its ends rather than its middle.**

answer: Force must pass through glue joints and the bolt hook, where contact areas are small and stress concentrates, while the middle carries only uniform tension. (4 points).

**9. (2 points) What do these tests show about where the structure needs material?**

Supplied synthetic observations:
Test A: Strut 15 cm deep, 3 mm square: bowed sideways at 9 kg.
Test B: Same strut with one mid-length brace: held 15 kg.
Test C: Chord 1.5 x 3 mm: never failed in any test.
Test D: Chord-to-bolt joint glued with a single drop: pulled off at 12 kg.

A. All members should be the same size

B. The strut needs bracing and joints need care; the chord is already strong enough

C. The chord needs to be much thicker

**Correct: B**

**10. (4 points) How does humidity affect a finished structure?**

Unit: Testing, failure analysis, and iteration.
Worked-case data: Your log shows: 9.2 g held 15 kg; 8.4 g held 15 kg; 7.9 g failed at 13.1 kg (strut buckled). What should the competition build be?
Separate investigation observations:
Builds 1-3: 8.1 g, 8.3 g, 8.2 g: all held 15 kg.
Builds 4-5: 7.4 g, 7.5 g: failed at 12 kg and 14 kg, strut buckled.
Build 6: 7.6 g with a stiffer, denser strut: held 15 kg.

answer: Wood absorbs moisture, adding mass and slightly reducing stiffness, so store structures dry and weigh them before competition. (4 points).

**11. (6 points) Solve this case and explain your reasoning: A chord joint must carry 300 N. Your glue holds about 2 N per mm^2 of long-grain overlap. How much bonded area do you need, with a safety factor of 1.5?**

reasoning: Required strength: 300 N x 1.5 = 450 N.
Area = 450 N ÷ 2 N/mm^2 = 225 mm^2.
A 15 mm x 15 mm overlap, or two gussets of about 113 mm^2 each, would do. (3 points).

conclusion: Joint area, not member size, decides whether the chord end survives. (3 points).

**12. (2 points) Define “Contact depth” in the context of Boomilever.**

answer: Vertical distance from the bolt centerline down to the wall contact. (2 points).

**13. (2 points) Define “Loading Assembly” in the context of Boomilever.**

answer: Block, eyebolt, chain, S-hook, and bucket; at most 1,500 g. (2 points).

**14. (2 points) Define “Testing Wall” in the context of Boomilever.**

answer: Plumb, rigid wall at least 40 x 30 cm holding the bolt. (2 points).

**15. (2 points) Define “Modulus of elasticity (E)” in the context of Boomilever.**

answer: Material stiffness; resistance to stretching or bending. (2 points).

**16. (2 points) Define “Pyramid layout” in the context of Boomilever.**

answer: Paired chords and struts that resist twisting and side loads. (2 points).

**17. (2 points) Define “Free-body diagram” in the context of Boomilever.**

answer: A sketch of one object with every external force acting on it. (2 points).

**18. (2 points) Define “Safety factor” in the context of Boomilever.**

answer: Ratio of capacity to expected load; margin against variation. (2 points).